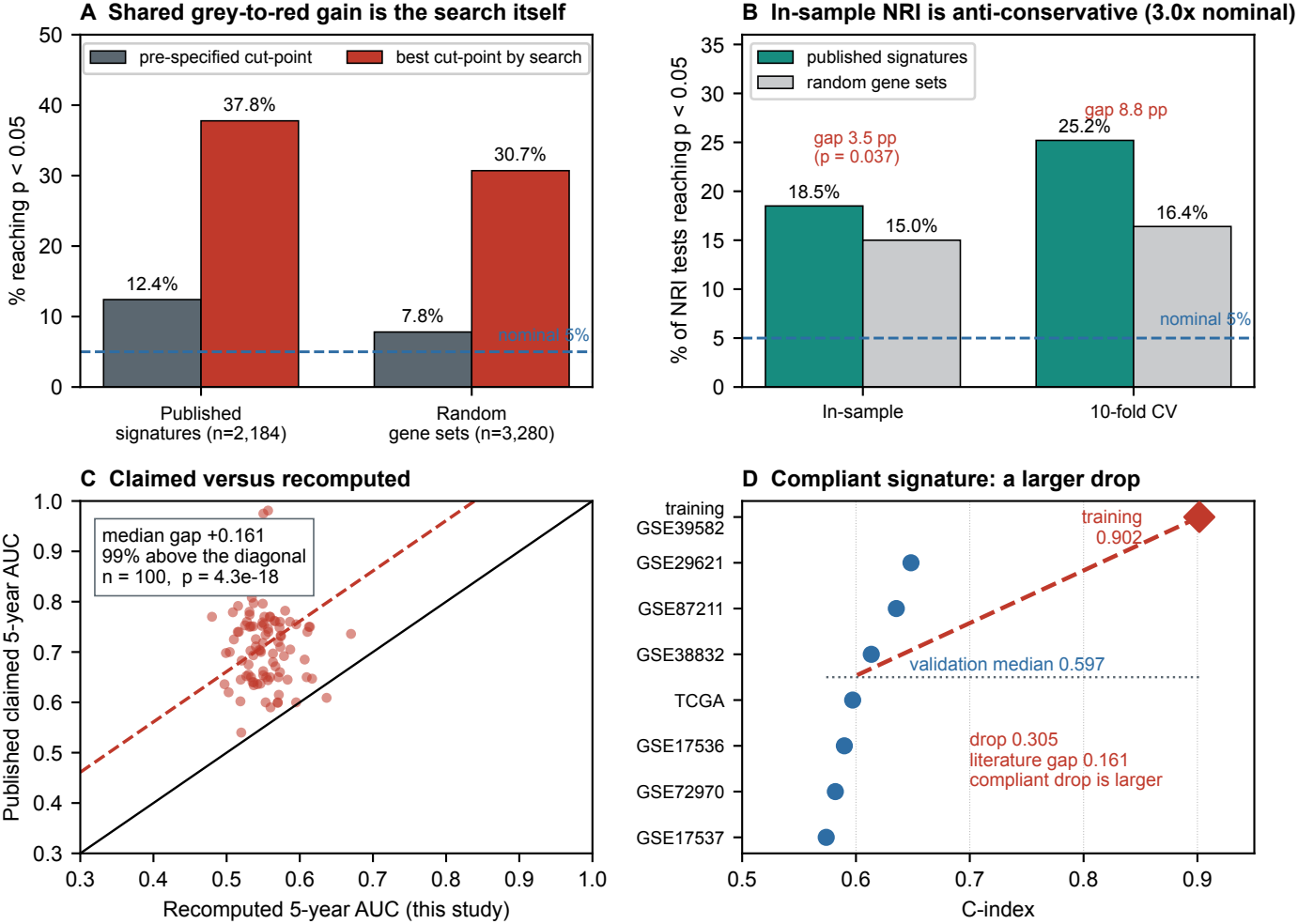
A Shared grey-to-red gain is the search itself
pre-specified cut-point
best cut-point by search
% reaching p < 0.05
12.4%
37.8%
7.8%
30.7%
nominal 5%
Published signatures (n=2,184)
Random gene sets (n=3,280)
B In-sample NRI is anti-conservative (3.0x nominal)
published signatures
random gene sets
% of NRI tests reaching p < 0.05
gap 3.5 pp (p = 0.037)
gap 8.8 pp
18.5%
15.0%
25.2%
16.4%
nominal 5%
In-sample
10-fold CV
C Claimed versus recomputed
median gap +0.161
99% above the diagonal
n = 100, p = 4.3e-18
Published claimed 5-year AUC
Recomputed 5-year AUC (this study)
D Compliant signature: a larger drop
training
GSE39582
GSE29621
GSE87211
GSE38832
TCGA
GSE17536
GSE72970
GSE17537
training 0.902
validation median 0.597
drop 0.305
literature gap 0.161
compliant drop is larger
C-index